STUDENT'S ID: _____ SIGNATURE_____

UNIVERSITY OF GHANA

(All rights reserved)

DEPARTMENT OF TEACHER EDUCATION

SCHOOL OF EDUCATION AND LEADERSHIP

COLLEGES OF EDUCATION

END OF SEMESTER ONE EXAMINATION FOR LEVEL 200, 2024/2025

B.ED. PROGRAMME

TEEG 209: THEORIES OF LEARNING MATHEMATICS FOR EARLY GRADE

(3 CREDITS)

TIME ALLOWED: 2 HOURS

Instruction: Answer **all** questions in Section A and any **three** questions in Section B.

Time: 2 hours

SECTION A

[25 Marks]

Answer all the questions in this section.

1. According to Bandura's social learning theory, which component is MOST essential for mathematical learning through observation?
   A. Physical capability to perform calculations
   B. Attention to the modeled behavior
   C. Previous success in mathematics
   D. Verbal instruction only

2. In a mathematics classroom applying social learning theory, what role should the teacher primarily assume?
   A. Lecturer who explains all concepts
   B. Model who demonstrates problem-solving strategies
   C. Observer who only assesses student work
   D. Administrator of tests and quizzes

3. Which scenario best exemplifies the application of social learning theory in mathematics education?
   A. Students working alone on practice problems
   B. Students watching videos of mathematical concepts
   C. Students observing and then imitating expert problem-solving strategies
   D. Students memorizing mathematical formulas

4. The concept of self-efficacy in mathematics learning primarily refers to:
   A. Actual mathematical ability
   B. Students' beliefs about their capability to succeed in mathematics
   C. Teacher's evaluation of student performance
   D. Parents' expectations of mathematical achievement

5. When applying social learning theory, peer modeling in mathematics is most effective when:
   A. The model is significantly more advanced than the observer
   B. The model is similar in ability level to the observer
   C. The model is younger than the observer
   D. The model is from a different academic background

6. The concept of vicarious reinforcement in mathematics learning suggests that:
   A. Students learn only through direct experience
   B. Students can learn from observing the consequences of others' actions
   C. Individual practice is the only way to master mathematics
   D. Teachers must always provide direct feedback

7. According to social learning theory, mathematical self-regulation develops through:
   A. Strict teacher control
   B. Observation and gradual internalization of standards
   C. Natural maturation
   D. Competition with peers

8. In terms of mathematical modeling, which statement best aligns with social learning theory?
   A. Models should only be perfect examples
   B. Models should show both successful and unsuccessful strategies
   C. Models should be limited to teacher demonstrations
   D. Models should focus only on final answers

9. The zone of proximal development in mathematical social learning refers to:
   A. The physical space where learning occurs
   B. The gap between what a learner can do alone and with assistance
   C. The maximum number of students in a class
   D. The time needed to master a concept

10. When applying social learning theory to mathematics homework, teachers should:
   A. Assign only individual work
   B. Provide worked examples and guided practice
   C. Give unlimited practice problems
   D. Focus solely on theoretical concepts

11. According to cognitive learning theory, which process is most crucial for mathematical concept formation?
    A. Memorization of formulas
    B. Schema construction and modification
    C. Repetitive practice
    D. Reward systems

12. In Piaget's theory, what characterizes the formal operational stage in mathematical thinking?
    A. Concrete manipulation of objects only
    B. Abstract reasoning and hypothetical thinking
    C. Pre-logical operations
    D. Sensorimotor experiences

13. Which cognitive strategy best supports problem-solving in mathematics?
    A. Rote memorization
    B. Metacognitive monitoring
    C. Speed calculations
    D. Emotional regulation

14. The concept of cognitive load in mathematics learning refers to:
    A. The number of problems assigned
    B. The amount of information processing required in working memory
    C. The physical weight of textbooks
    D. The duration of mathematics classes

15. Which instructional approach aligns best with cognitive learning theory?
    A. Direct instruction only
    B. Scaffolded problem-solving
    C. Unlimited practice problems
    D. Competitive assessments

16. According to cognitive load theory, which approach is most effective for teaching complex mathematical concepts?
    A. Presenting all information simultaneously
    B. Breaking down information into manageable chunks
    C. Using pure discovery learning
    D. Focusing on memorization

17. The concept of "meaningful learning" in mathematics implies:
    A. Learning only practical applications
    B. Connecting new information to existing knowledge
    C. Memorizing formulas
    D. Practicing procedures repeatedly

18. In cognitive theory, mathematical problem-solving skills are best developed through:
    A. Repeated practice of similar problems
    B. Development of strategic thinking and metacognition
    C. Memorization of procedures
    D. Teacher demonstration only

19. Working memory limitations in mathematics learning suggest that teachers should:
    A. Teach as many concepts as possible in one lesson
    B. Present information in small, organized units
    C. Focus only on drill and practice
    D. Avoid using visual aids

20. Which cognitive strategy best supports mathematical concept formation?
    A. Silent individual practice
    B. Multiple representations and connections
    C. Memorization of rules
    D. Speed-based exercises

21. According to behaviorist theory, what is the most effective way to teach new mathematical concepts?
    A. Breaking complex tasks into smaller, sequential steps
    B. Presenting abstract theories first
    C. Group discussion
    D. Discovery learning

22. In behaviorist learning theory, what role do mistakes play in mathematics learning?
    A. They are opportunities for creative thinking
    B. They should be immediately corrected to prevent reinforcement
    C. They are necessary for cognitive development
    D. They should be ignored completely

23. Which type of reinforcement would a behaviorist recommend for mathematics learning?
    A. Delayed feedback on weekly tests
    B. Immediate feedback on each problem
    C. Peer evaluation
    D. Self-assessment

24. The behaviorist concept of 'shaping' in mathematics instruction refers to:
    A. Drawing geometric shapes
    B. Gradually reinforcing closer approximations to desired behavior
    C. Modeling problem-solving
    D. Grouping students by ability

5. According to behaviorist theory, what is the most important factor in mathematics success?

A. Environmental conditions and reinforcement
B. Innate ability
C. Emotional state
D. Social interaction

Section B [75 Marks]

Answer any **THREE** questions in this Section in the answer booklet provided.

In each question, show working in the answer booklet provided, including the answer.
Note: ***If you answer more than three questions, only the first three questions would be marked.***

1.	a. How will you distinguish *a learning* from a *learning theory*	**8 Marks**
	b. Assign five (5) reasons why a professional mathematics teacher should bother himself/herself with the knowledge of *learning theories*	**15 Marks**
	c. How are the *developmentalist theory of learning* different from the *cognitivist theory of learning different*?	**2 Marks**
2.	*a.* What are the two most Important types of learning in *behavioral theory*	**3 Marks**
	b. Explain how learning occurs under each of the types as identified in (2a)	**12 Marks**
	c. Describe three implications of the of *cognitivist Learning Theory* to the teaching of mathematics	**9 Marks**
	a. what is the main focus for the cognitivist approach?	**5 Marks**
	b. Explain two major cognitivist factors that influences the thinking of a mathematics student?	**10 Marks**
	c. Describe three implications of the of *cognitivist Learning Theory* to the teaching of mathematics	**10 Marks**
3.	a. what is the main focus for the constructivist approach?	**2 Marks**
	b. What are the most two important concepts of constructivist learning theories?	**4 Marks**
	c. Explain these two concepts as identified in (4a) to the teaching and	**10 Marks**

	learning of mathematics.	
	d. Describe three implications of the of social learning *Theory* to the teaching of mathematics	**9 Marks**
5.	a. How is learning perceived in social learning?	**4 Marks**
	b. Briefly explain the stages in experiential learning as described by Kolb	**12 Marks**
	c. Describe three implications of the of *Behaviourist Learning Theory* to the teaching of mathematics,	**9 Marks**